Lab: test a claim

Session 2 — Data management

Data Analysis in R using Agentic AI Assistants · Kevin Koehler · Friday 30 October 2026

Since the end of the Cold War, internal armed conflict has been concentrated in poorer countries.

Is that true? Work in **pairs**. You will go through five steps. The assistant helps with Steps 2 and 5. Everything else is your job.

Keep notes as you go. Create a file `notes/lab2.md` in your project and write in it: your answers to the questions below, every prompt you send, and anything that went wrong. You will need these notes at the end of class and again next week.

Step 1: Find the data

about 20 min

What you do

Use these two sources:

- **Conflict:** the Uppsala Conflict Data Program (**UCDP**), `https://ucdp.uu.se`. Use the **UCDP/PRIO Armed Conflict Dataset**, which records armed conflicts year by year.
- **Income:** the **World Bank**'s World Development Indicators, `https://data.worldbank.org`. Use **GDP per capita**.

Find the latest version of each, download it into your project's `data/` folder (CSV is easiest), and find and open the **codebook** or indicator description.

No AI in this step. Use a search engine, the producers' websites, and the codebooks. An assistant will happily describe datasets and variables that do not exist.

For each dataset, fill in this table (copy it into your notes):

	UCDP	**World Bank**
Exact dataset or indicator name, version		
File you downloaded		
One row = ? (e.g. one event, one conflict in one year, one country in one year)		
How is a country identified? Name? A code? Which code system?		
Which years are covered?		
Which variable will you use? How exactly is it measured? In what unit?		
Is the file **wide** (one column per year) or **long** (one row per year)?		
How is "missing" coded?		

Look at the codebook, not only at the website summary. UCDP's definition of "armed conflict" (how many deaths? who must be fighting?) and its "type of conflict" variable decide what you are counting as *internal*. The World Bank offers several versions of GDP per capita (current US$, constant US$, PPP): which one fits a comparison across countries and years?

Step 2: Prompt the merge

about 10 min

What you do

Write a prompt that tells the assistant to combine your two datasets into **one table with one row per country and year**, and to save it as `output/conflict_income.csv`. Start a fresh OpenCode session first (`/new`).

Your prompt has to say:

- which **files** to use, and where they are;
- which column **identifies the country** in each file, and which **code system** it uses;
- which **years** you want;
- what **one row** of the final table should be.

Before you send it, look back at your Step 1 table and ask:

- Do both datasets identify countries **the same way**? If not, what has to happen first?
- Is each dataset **already** one row per country-year? If not, what has to happen first?
- Is each dataset wide or long?

If the answer to any of these means extra work, your prompt must say so. Copy the prompt into your notes.

Step 3: Inspect the data

about 15 min

What you do

Load it in the Console with `d <- read.csv("output/conflict_income.csv")` and look at it with `View(d)`. Answer the questions below by looking at the data and typing short commands in the **Console**, not by asking the assistant.

Check — without asking the assistant

1. **Did the merge work?** How many rows? How many countries? How many years? Is the number of rows close to countries × years?
   Useful: `nrow(d)`, `length(unique(d$country))`, `range(d$year)`.
2. **What did not match?** Which countries are in one dataset but not the other? Write down at least three. Why those?
3. **Is it tidy?** Is there exactly one row per country-year? Does any country-year appear twice? Does any cell contain two values?
   Useful: `d %>% count(country, year) %>% filter(n > 1)`
4. **Is everything there?** Take a country that had no conflict in a given year. Is it in the table? Is its conflict variable `0`, or `NA`? Which one *should* it be for our claim?
5. **Spot-check by hand.** Pick two countries you know something about. Do their values look right for three different years? Compare with the original files.

Found a problem? Write a **second prompt** that fixes it, then check again. Note in `notes/lab2.md` what was wrong and how you noticed.

Step 4: Read the code — about 10 min

What you do

Open the script the assistant wrote. Read it with your partner. You should be able to explain every step to someone else.

1. Find the step that **converts or matches country identifiers**. From what to what? What happens to countries it cannot convert? Did it tell you?
2. Find any step that **reshapes** (wide to long) or **aggregates** (several rows into one). What is one row before that step, and after?
3. Find the **merge** (a `*_join()`). Which kind of join? What happens to rows that find no match?
4. Pick the line you understand **least**. Look up the function with `?name` in the Console. Write in your notes, in your own words, what it does.

Step 5: Design the test — about 15 min

What you do

Decide how to test the claim **before** you prompt anything. Write your answers in your notes. Then write a prompt that implements your test, and ask for a table or a chart you can read, not just one number.

1. What does **"concentrated"** mean? For example: conflict is more common among poorer countries; or the average income of country-years with conflict is lower than of those without. Choose one and say why.
2. What does **"poorer"** mean? Income as a number? Groups (e.g. four groups from poorest to richest)? Relative to what?
3. Which **years** count as "since the end of the Cold War"?
4. Could the pattern **run the other way**? War destroys income. If a country is poor *during* a conflict, does that support the claim? Which year's income should you use?
5. What result would make you say the claim is **false**?

At the end of class

Be ready to answer, as a pair:

- Is the claim true? How sure are you?
- Which of your decisions changed the answer the most?
- Where did the assistant make a decision you had not asked for?

Keep `data/`, `output/conflict_income.csv`, the script, and `notes/lab2.md`. Add one line to `log.md`: something the assistant got wrong today, and how you caught it.

Reminder: what we want, and what we check

Tidy data: each variable a column, each observation a row, each value a cell. What "one observation" is depends on your unit of analysis.

Country codes: the same country can appear under several names and several code systems: Correlates of War (COW), Gleditsch & Ward (G&W), ISO 3166 three letters (ISO-3c, e.g. `TUN`) or numbers (ISO-3n, e.g. `788`). The R package `countrycode` converts between them. Always check what it could *not* convert.

After every merge: rows before and after — should the number change? Is the key unique in each table? Which rows found no match, and why?

After every reshape: count the rows (countries × years); is `year` a number or text?